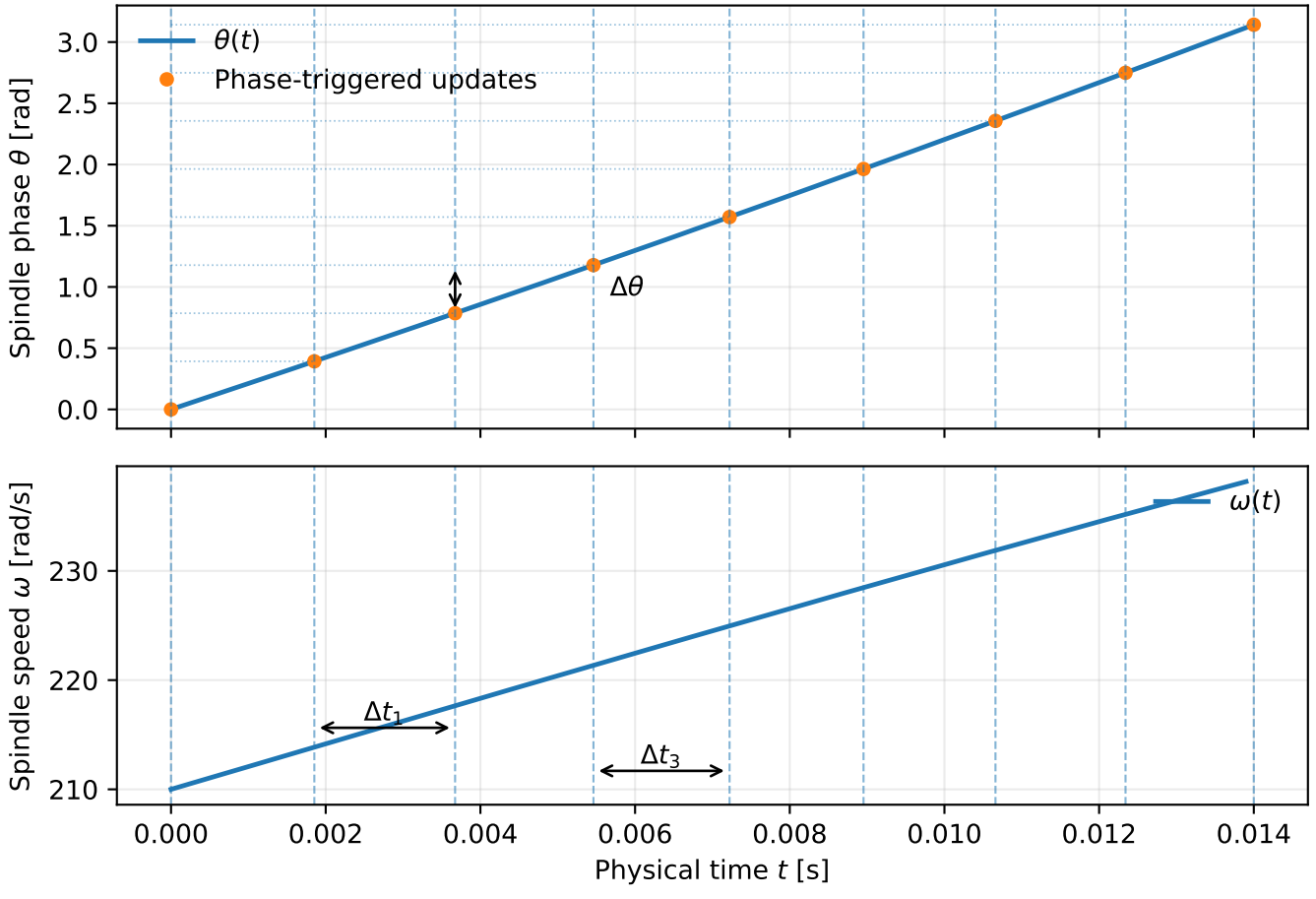
Spindle-phase-synchronous MHE–NMPC updating
θ(t)
Phase-triggered updates
Δθ
Spindle phase θ [rad]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
ω(t)
Δt1
Δt3
Spindle speed ω [rad/s]
210
220
230
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
Physical time t [s]